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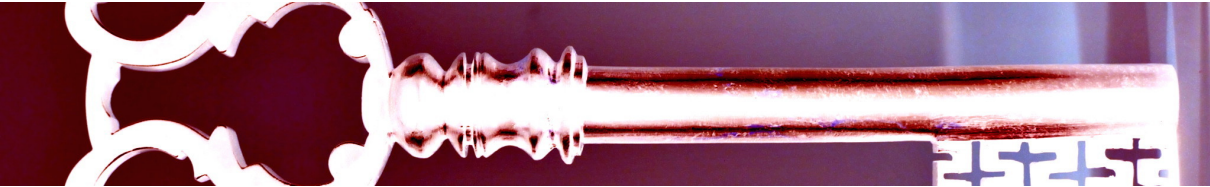

Probability, Information, and Coding Theory

Cryptography

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12 March 2026

Randomness

Philosophical questions

- Epistemology
- Quantum physics
- Determinism (NOT: non-determinism in theoretical CS)
- Related: frequentist vs Bayesian

Pragmatic uses

- Model missing information
- Represent excess complexity with simplified models
- Introduce by sampling
  $\rightarrow$ approximate answers at lower measurement effort
- Introduce to enforce independence
  $\rightarrow$ advantage by “choice” of secret parameters (e. g., keys)

Agenda

1. **Probability theory**
2. Information theory
3. Coding theory

Probability

A **probability space** is a mathematical construct to model real-world phenomena (or “experiments”) with **random** states.

It is specified by three components:

1. **Sample space** Ω: set of all possible states (outcomes of the experiment)
2. **Event set** $\mathcal{F}$: set containing subsets of Ω (events)
3. **Probability function** P: map that assigns a probability value to each event in $\mathcal{F}$

Probability Axioms

Three properties hold for probability functions:

1. $P(E) \geq 0, \forall E \in \mathcal{F}$: every event E has non-negative probability.
2. $P(\Omega) = 1$: the set of all possible outcomes has probability 1.
3. $E_i \cap E_j = \emptyset \quad \Rightarrow \quad \forall\,(i,j), i \neq j, P\left(\bigcup_i E_i\right) = \sum_i P(E_i)$: for any countable sequence of *disjoint* (incompatible) events, the probability of their union is the sum of their probabilities.

Useful **corollaries:**

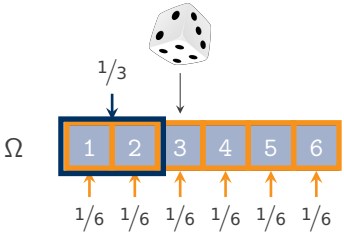
- $P(\emptyset) = 0$: the empty event has always zero probability.
- $P(E) \leq 1 \quad \forall\, E \in \mathcal{F}$: the maximum probability of each event is 1.
- $A \subseteq B \quad \Rightarrow \quad P(A) \leq P(B)$: events contained in other ones have lower probability.

Example

Experiment: tossing a fair dice with six numbered sides. Every toss is a trial.

Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Event set: $\mathcal{F} = 2^{\Omega}$, power set of the sample space.



1. Initial assumption: the dice is fair $\Rightarrow P(\{1\}) = P(\{2\}) = \ldots = P(\{6\})$
2. Second axiom: $P(\Omega) = P\left(\bigcup_{i=1}^{6}\{i\}\right) = 1$
3. Third axiom: $\sum_{i=1}^{6} P(\{i\}) = P\left(\bigcup_{i=1}^{6}\{i\}\right) = 1$
4. From 1, 2, and 3: $P(\{i\}) = 1/6, \forall i$
5. From 4 and second axiom: $P(\{i\} \cup \{j\}) = 1/6 + 1/6 = 1/3$

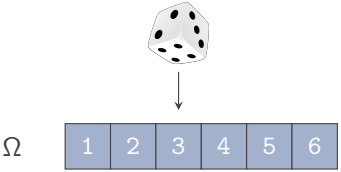
   ...

Random Variables

A **random variable** X is related to a random experiment and its **realization** x observed from a trial.

It is a mapping from the sample space Ω to a **domain** $\mathcal{X}$.

Examples:



- X_1 is the side number: $\mathcal{X}_1 = \{1, 2, 3, 4, 5, 6\}$
- X_2 is half of the side number: $\mathcal{X}_2 = \{0.5, 1, 1.5, 2, 2.5, 3\}$
- X_3 is 0 if the side number is even, 1 if it is odd: $\mathcal{X}_3 = \{0, 1\}$. After a trial, we get:

$$1, 3, 5 \longrightarrow \qquad x_3 = 1$$

or

$$2, 4, 6 \longrightarrow \qquad x_3 = 0$$

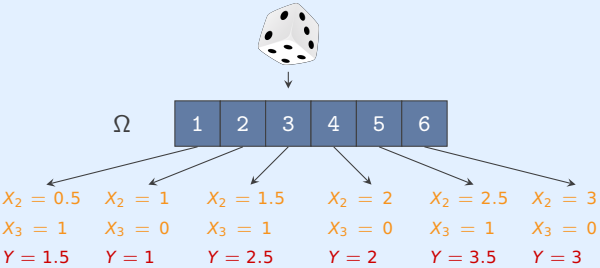
Random Variables (cont'd)

For discrete domains, probability functions $P(X = x)$ are defined directly on the domain $\mathcal{X}$. For continuous domains, $P(X = x)$ is not available. $P(X \leq x)$ is studied instead.

Random variables can be **algebraically combined** to obtain other random variables (depending on the domain and scale level of the measurement).

Example

- X_2 is half of the side number, X_3 is 0 if the side number is even and 1 if it is odd.
- $Y = X_2 + X_3$



Conditional Probability and Independence

Now consider **two** random variables X and Y.

The **conditional probability** that $X = x$ given that $Y = y$ is

probability that $X = x$
when knowing that $Y = y$

$$P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

joint probability that $X = x$ AND $Y = y$

marginal probability that $Y = y$

Definition

Two random variables X and Y are **statistically independent** iff

$$P(X = x \mid Y = y) = P(X = x) \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y} .$$

The probability that $X = x$ does not change when knowing the realization of Y !

Equivalently, the joint probability is the product of the marginal probabilities

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y).$$

Bayes' Theorem

Bayes' theorem tells us how to swap the conditions

$$P(X = x \mid Y = y) = \frac{P(Y = y \mid X = x) \cdot P(X = x)}{P(Y = y)}$$

if we know the **marginal distributions** $P(X = x)$ and $P(Y = x)$.

Bayesian learning

After observing $Y = y$, Bayes' theorem allows us to update the **prior** $P(X = x)$ to obtain the **posterior** $P(X = x \mid Y = y)$.

NB: Ignoring the general information $P(X = x)$ in favor of the specific information $P(Y = y \mid X = x)$ is a common behavioral bias called **base rate fallacy**.

Example

Catch attackers by sanctioning false password entries?
Consider a typical password dialogue. Attackers enter a wrong password with probability 99.999 %. Authorized parties enter a wrong password in one out of ten cases.
About 2 % of the population are attackers.

A system detects a wrong password, how likely is an attacker trying to log in?

Wrong approach: for attackers, the password is wrong in 99.999 % of cases, so it is highly probable that it's an attacker.
Base rate fallacy: We are not considering that attackers are only 2 % of users.

Correct approach: applying Bayes' theorem

$$P(\text{attacker} \mid \text{wrong}) = \frac{P(\text{wrong} \mid \text{attacker}) \cdot P(\text{attacker})}{P(\text{wrong})} = 0.1695$$

$$P(\text{wrong} \mid \text{attacker}) = 0.99999 \tag{1}$$

$$P(\text{wrong} \mid \overline{\text{attacker}}) = 0.1 \tag{2}$$

$$P(\text{attacker}) = 0.02 \quad \Rightarrow \quad P(\overline{\text{attacker}}) = 0.98 \tag{3}$$

$$P(\text{wrong}) = P(\text{wrong} \mid \text{attacker}) \cdot P(\text{attacker}) + P(\text{wrong} \mid \overline{\text{attacker}}) \cdot P(\overline{\text{attacker}}) = 0.118 \tag{4}$$

Distribution Models

Distribution models simplify $P(X = x)$, indicated by $P_X(x)$, to classes of functional forms, generally specified by a vector of **parameters** θ.

Discrete case: $\mathcal{X}$ is a discrete set.

- **Bernoulli** distribution

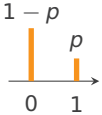
  X is a binary random variable, e. g.,
  $\mathcal{X} = \{0, 1\}$,
  that depends only on the scalar parameter
  $p \equiv \theta$ such that:

  $$P_X(1) = p \qquad P_X(0) = 1 - p$$

- **Binomial** distribution: Y is the sum of n independent Bernoulli r. v. with parameter p. $P_Y(k)$ is the probability of observing $X = 1$ exactly k times out of n. $\theta = (n, p)$

- . . . (many more)

Trial 1



⋮ ⋮

Trial n

$$Y = X_1 + \cdots + X_n$$

Relations to Cryptography

1. Cryptographic security is limited by the luck of attackers to guess the correct key.
2. Security definitions specify the event of attacker success.
3. Security proofs show that this event has “sufficiently low” probability.
4. Conditional probabilities can incorporate observations of the attacker related to its goal, e. g., other plaintext–ciphertext pairs produced by the system under attack.
5. Distribution assumptions make highly non-linear functions tractable for analysis.

Relations to Cryptography (cont'd)

Limitations — why probability theory isn't enough

6. Information theory generalizes probability theory to sequences of random variables.
7. Coding theory extends information theory to deal with biases in the probability distributions of natural sources (e. g., English plaintext), and with adding artificial bias for transmission over noisy communication channels.
8. Sometimes the attacker's success probability is close to one, but goes down if we assume that the attacker cannot perform some expensive operations. This is where we need number theory and complexity theory.
9. Probability theory cannot deal with strategic (re)action. We study the (ε-)worst case or—not in this lecture—resort to game theory by making (restrictive) assumptions about the attacker's behavior.

Agenda

1. Probability theory
2. **Information theory**
3. Coding theory

Information Theory

What is information?

Conventions

In information theory, discrete random variables are also called sources and their domains alphabets.

Memoryless is a shorthand for independent and identically distributed.

Consistent with before, we use

- calligraphs $\mathcal{X}$ for alphabets (domains),
- capital letters X for sources (random variables) and the corresponding
- lower case letters x for realizations.

We distinguish variables X from discrete symbols $\mathtt{X}$.

Shannon Entropy

Let X be a memoryless source over $\mathcal{X} = \{x_1, x_2, \ldots, x_{|\mathcal{X}|}\}$ with distribution $P_X(x)$.

Entropy $H(X)$ is defined as

$$H(X) = -\sum_{i=1}^{|\mathcal{X}|} P_X(x_i) \log_b P_X(x_i).$$

The unit depends on base: bit ($b = 2$), nat ($b = e$), ban ($b = 10$)

- Entropy is a measure of disorder and uncertainty of a source.
- It is based on the **self-information**

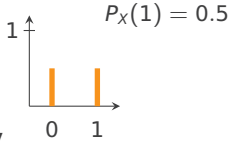
$$I(x) = -\log_b P_X(x),$$

  which measures the surprise of observing $X = x$.
- Entropy is the **expected self-information** under P_X.

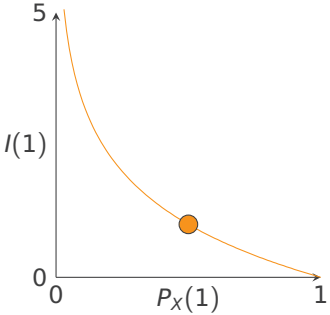
Claude Shannon, *A Mathematical Theory of Communications*, 1948.

Visualization of a Special Case

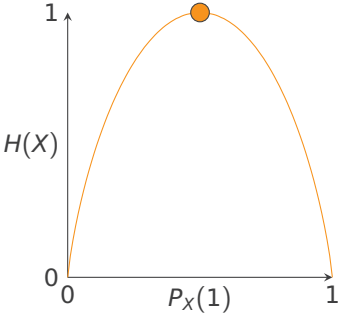
Binary source: $\mathcal{X} = \{0, 1\}$, $b = 2$



Self-information



Shannon entropy



Maximum $H(X) \Leftrightarrow P_X(0) = P_X(1) = 1/2 \Leftrightarrow$ uniform distribution over $\mathcal{X} = \{0, 1\}$

$\rightarrow$ This generalizes to alphabets with arbitrary cardinality $|\mathcal{X}|$.

Multiple Sources

Let $X \overset{P_X}{\rightsquigarrow} \mathcal{X}$ and $Y \overset{P_Y}{\rightsquigarrow} \mathcal{Y}$.

- **Joint entropy**

$$H(X,Y) = -\sum_{i=1}^{|\mathcal{X}|}\sum_{j=1}^{|\mathcal{Y}|} P_{X,Y}(x_i,y_j)\log_b P_{X,Y}(x_i,y_j)$$

$H(X,Y) = H(X) + H(Y) \Leftrightarrow X$ and Y are statistically independent.

- **Conditional entropy** (equivocation)

entropy of the conditional distribution

$$H(X \mid Y) = \sum_{j=1}^{|\mathcal{Y}|} P_Y(y_j)\cdot H(X|Y=y_j)$$

$$= H(X,Y) - H(Y) \quad \text{“chain rule”}$$

Mutual Information

The **mutual information** $I(X;Y)$ is defined as:

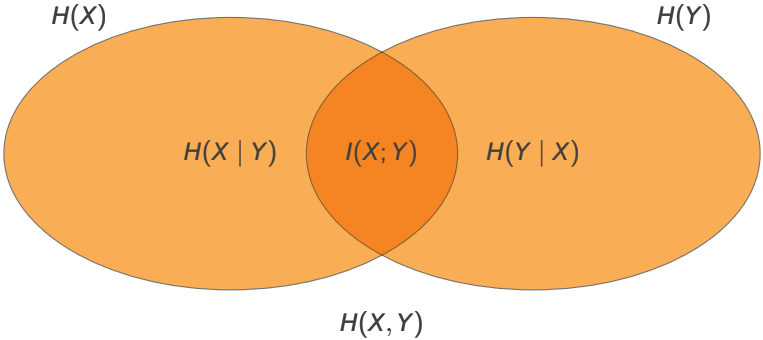
$$I(X;Y) = H(X) + H(Y) - H(X,Y).$$

It is a measure of mutual dependence between X and Y.

Properties:

- non-negativity: $I(X;Y) \geq 0$
- symmetry: $I(X;Y) = I(Y;X)$
- $I(X;Y) = 0 \Leftrightarrow X$ and Y are statistically *independent*

Visualization



Interpretations of mutual information $I(X;Y)$

- The amount of information that X and Y share;
- How much we learn about X by observing Y, and vice versa.

Agenda

1. Probability theory
2. Information theory
3. **Coding theory**

Redundancy

Informally:

$$\begin{aligned}\text{redundancy} &= \text{max. entropy over } \mathcal{X} - H(X) \\ &= \log_b |\mathcal{X}| - H(X)\end{aligned}$$

[Unit: bits (for $b = 2$)]

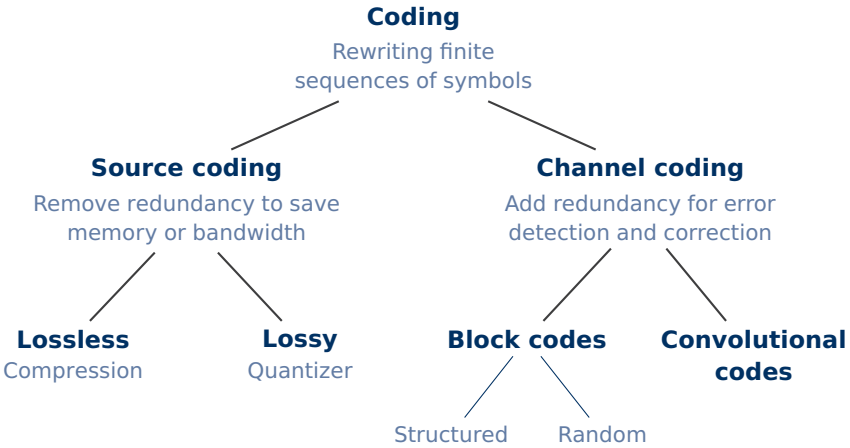
Or in terms of symbols:

$$\text{redundancy} = \text{symbols used} - \text{symbols needed}$$

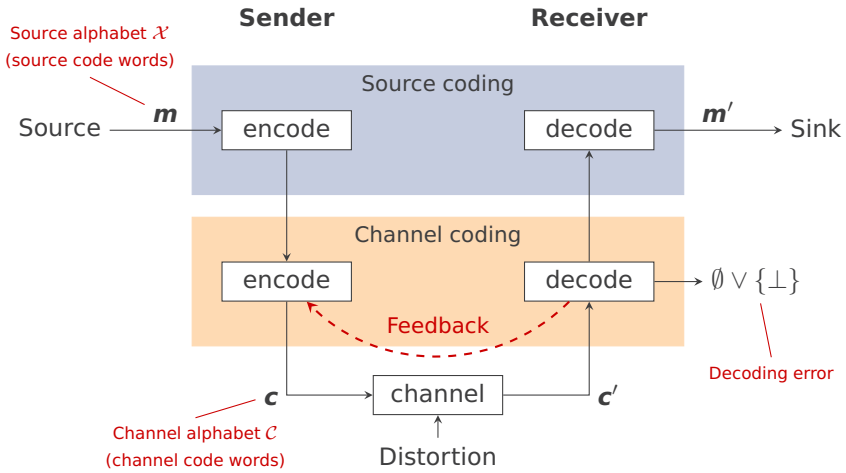
[Unit: symbols]

Redundancy bounds the savings of data compression from above.

Coding is Not Cryptography



Coding in Communication Systems



Source Coding

Objective

Remove redundancy of a message in order to save, e.g., memory.

Example: **Run-length encoding (RLE)** and variants. $\mathcal{X}$ has all 8-bit ASCII characters.

`AAAAAAACCAAAAAAABBBBBBBBBDQQQQQQQQYXZZQLKKKLKHZMMN` (50 bytes)

18 segments, 10 with 1 letter

Plain RLE

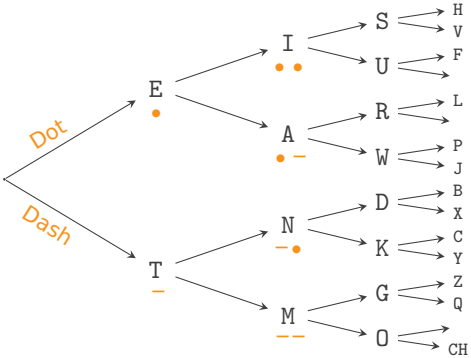
Char.	Deci.	Enc.	#bytes
A	65	07 65	2
C	67	02 67	2
A	65	07 65	2
⋮	⋮	⋮	⋮
M	77	02 77	2
N	78	01 78	2
			36

Optimized RLE

Enc.	#bytes
65 65 07	3
67 67 02	3
⋮	⋮
68	1
⋮	⋮
77 77 02	3
78	1
	34

Source Coding

Old-world example: **Morse's code**



Variable Length *D*-ary Prefix Codes

Definition

A **variable length *D*-ary code** is a source code where the codewords are strings of symbols from an alphabet with cardinality D that can have different length.

Examples: The Morse code is a 3-ary code created with symbols $\bullet$, $-$, and a separator

Definition

If no valid codeword is a prefix (i.e., start) of any other valid codeword, then the codewords have the **prefix property**.

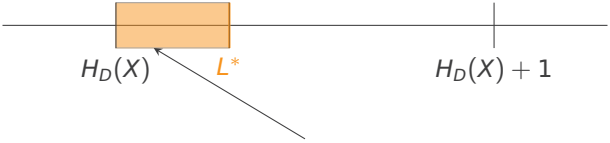
Examples: The codewords $\{\mathtt{A}, \mathtt{BA}, \mathtt{BB}\}$ have the prefix property.
The codewords $\{\mathtt{A}, \mathtt{AB}, \mathtt{BB}\}$ do not.

Theoretical Bounds

Source coding theorem

Let L^* be the minimum expected length for any variable length D-ary prefix code of a source $X \overset{P_X}{\sim} \mathcal{X}$. Then,

$$H_D(X) \leq L^* < H_D(X) + 1$$



This coding loss goes down if $P_X(x_i)$ are close to inverse powers of D.

Huffman codes are optimal prefix codes for a memoryless source with distribution P_X.

(But not the optimal form of lossless data compression for any distribution P_X.)

Channel Coding

Objective

Add redundancy to a message in order to reduce the impact of transmission errors.

Real-world example: Spelling alphabets

A	Alpha
B	Bravo
C	Charlie
D	Delta
E	Echo
⋮	⋮
Y	Yankee
Z	Zulu

Terminology for Linear Block Codes

Linear block codes are characterized by the tuple

length: number of symbols per channel codeword

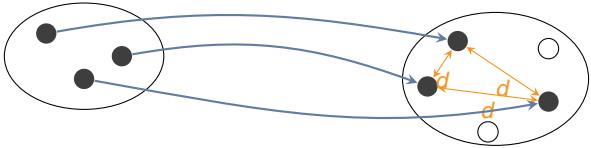
minimum (Hamming) **distance** between any two channel codewords

$$[n, k, d]_q$$

rank: number of symbols per source codeword

q-**ary code:** alphabet with q symbols used to create codewords

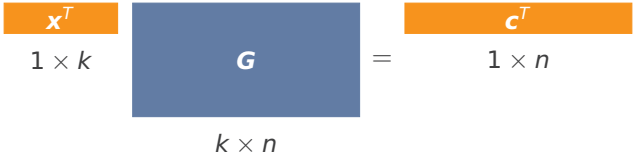
Source codewords (k-dimensional space)



Channel codewords (n-dimensional space)

Linear Block Codes In A Nutshell

A source codeword $\boldsymbol{x}$ is encoded into the channel codeword $\boldsymbol{c}$ through the multiplication, typically in GF(2), with the **code generator matrix** $\boldsymbol{G}$:



For every $\boldsymbol{G}$, there exists a **parity check** matrix $\boldsymbol{H}$ such that $\boldsymbol{c}$ is a valid channel codeword iff

$$\boldsymbol{H}\boldsymbol{c} = \boldsymbol{0}.$$

When an n-dimensional codeword $\boldsymbol{c}$ is received, the vector $\boldsymbol{s} = \boldsymbol{H}\boldsymbol{c}$ (called **syndrome**) is computed: if $\boldsymbol{s} \neq \boldsymbol{0}$, an error is detected.

A linear block code can **detect** $d-1$ symbol errors per codeword and **correct** $\lfloor \frac{d-1}{2} \rfloor$ of them with the closest valid codeword.

Example: Hamming Codes

Hamming code: $\forall i \in \{2, 3, \dots\}\ \exists$ a linear code with parameters

$$[2^i - 1, 2^i - i - 1, 3]_2$$

Example: $i = 3 \Rightarrow [7, 4, 3]_2$

$\boldsymbol{x}^T = (1, 0, 1, 0)$, $\boldsymbol{c}^T = \ ?$

$$\boldsymbol{G} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Hamming 1950

Example: Hamming Codes

Hamming code: $\forall i \in \{2, 3, \dots\}\ \exists$ a linear code with parameters

$$[2^i - 1, 2^i - i - 1, 3]_2$$

Example: $i = 3 \Rightarrow [7, 4, 3]_2$

$$\boldsymbol{x}^T = (1, 0, 1, 0),\ \boldsymbol{c}^T = \ ?$$

Decoding:

received sequence	decoder output
$(1, 0, 1, 0, 0, 0, 1)$	
$(1, 0, 1, 1, 0, 0, 1)$	
$(1, 1, 1, 0, 0, 0, 0)$	
$(1, 1, 1, 0, 1, 0, 0)$	

Hamming 1950

Need to Know: Information and Coding Theory

We have seen that . . .

- Entropy is a “measure of disorder”.
- Coding does not mean cryptography.
- Codes with prefix property resolve ambiguity among variable-length messages.
- Every finite sequence of symbols can be encoded in a (shorter) sequence, and the optimal expected length is bounded by the entropy of the source.

Please reflect on what you have learned in order to argue that . . .

- It is hard to reach the minimum for empirical sources with memory.
- For every channel with strictly positive mutual information between input and output, a code can be constructed that bounds the bit error rate from above.

Relations to Cryptography

1. Next week, we will use conditional entropy to measure the security of symmetric encryption.
2. We don't want to make assumptions about the plaintext distribution. Redundancy in the plaintext may help the attacker to know when s/he has guessed the correct key.
3. The prefix property helps avoiding ambiguity of message authentication.
4. Error correction may support availability.
5. Real-world communication systems may need to combine source coding, channel coding, encryption, and message authentication, ensuring that no component defeats the purpose of any other component.

Further Reading

- T. M. Cover & J. A. Thomas (2006): Elements of Information Theory, 2nd edition, Wiley, New York, 776 pages.
- H. Klimant, R. Piotraschke & D. Schönfeld (2012): Informations- und Kodierungstheorie, 4. Auflage, Teubner, Wiesbaden, 296 pages.

Syllabus

Summer Term 2026

	05.03.26	1.	Introduction, basic terminology	
	12.03.26	2.	Probability, information, and coding theory for cryptography	
	19.03.26	3.	Symmetric encryption, perfect secrecy	
Tue	24.03.26	4.	Block ciphers	HSB 9
	16.04.26	5.	Cryptanalysis of classical ciphers	
	23.04.26	6.	Number theory for cryptography	
	30.04.26	7.	Computational complexity for cryptography	
	07.05.26	8.	Asymmetric encryption	
	21.05.26	9.	Message authentication	
	28.05.26	10.	Elliptic curve cryptography	
	11.06.26	11.	Commitments, zero-knowledge, and secret sharing	
Tue	23.06.26	12.	Q & A for the exam	HSB 9
	25.06.26		**Written exam**	

Last update: March 12, 2026